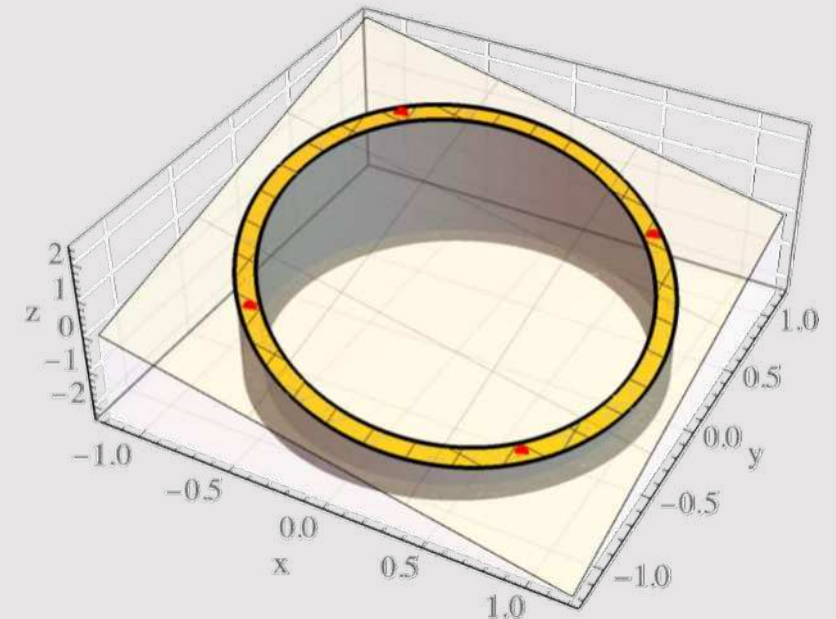
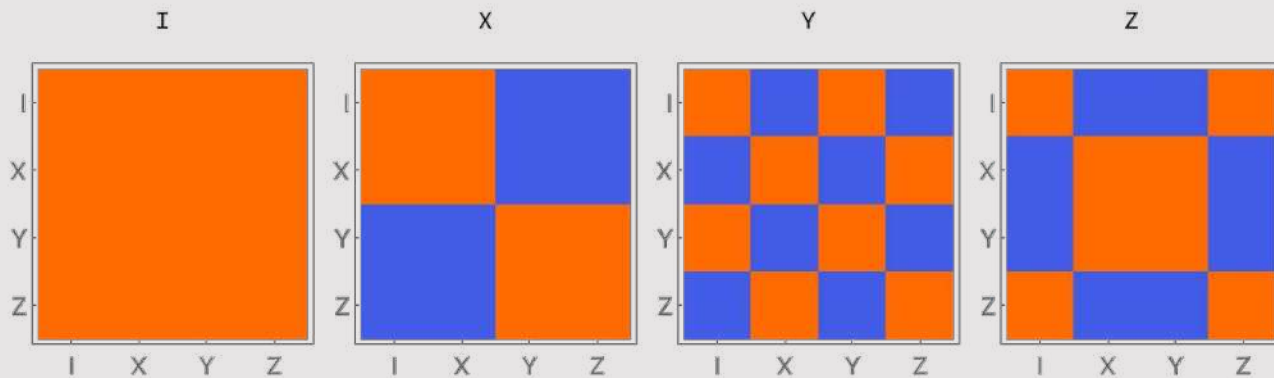
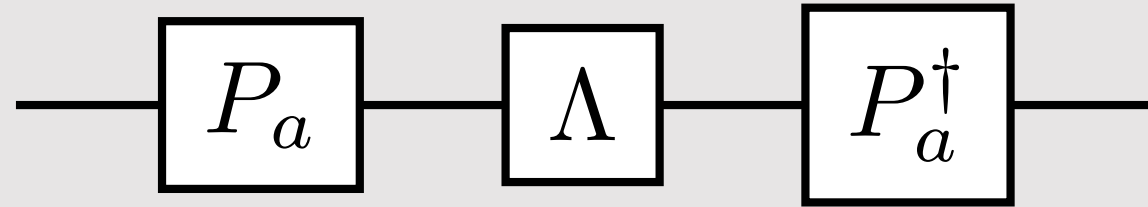


Primer on Pauli Twirling



Zlatko Minev

2022-04-20, 07-11

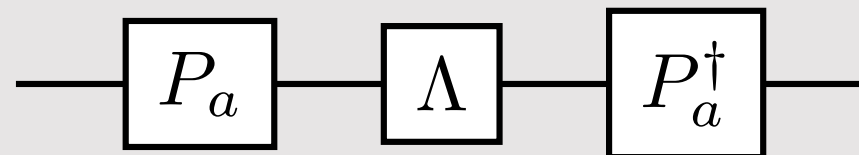
Twirling 101: Overview

Twirl operationally

Simple example

General application

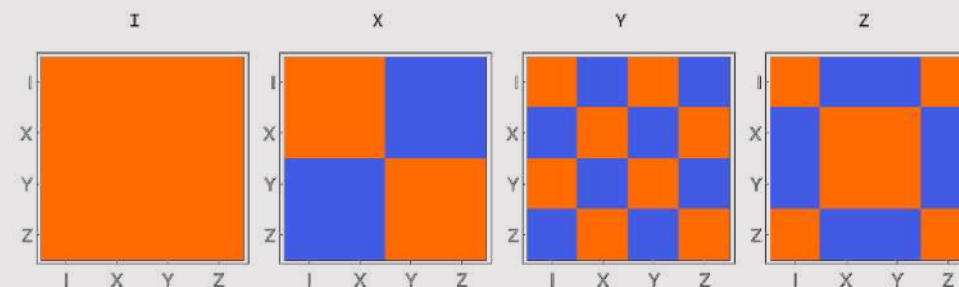
Summary



Theory of twirling

Why does twirling work?

Masking channels

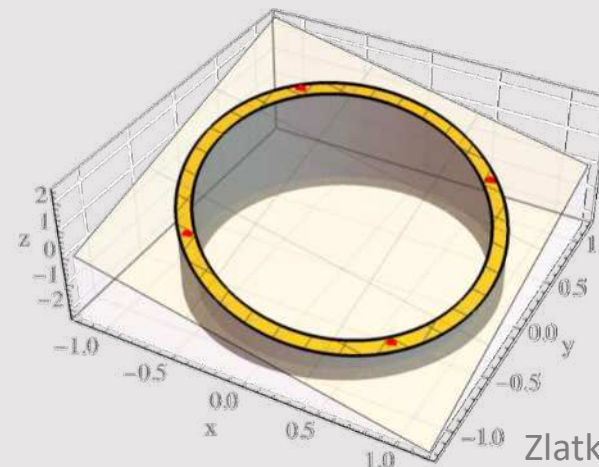


Optional: Advanced

Why is the Pauli group special for twirling?

Other twirl groups

Designs





* pikisuperstar

Refresher

More general

Pauli gates & mixed states

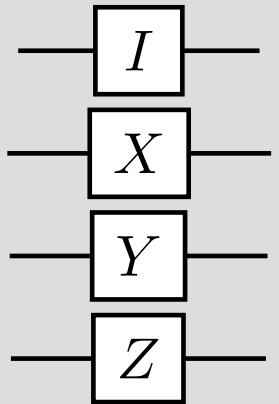
Single-qubit Pauli gate

Pauli gate on a mixed state: conjugation of ρ by Pauli

$$\rho \text{ --- } \boxed{P_a} \text{ --- } \mathcal{P}_a \rho = P_a \rho P_a^\dagger$$

Pauli gate

Single-qubit Pauli set



$$P_a \in \mathbf{P}$$

$$\mathbf{P} := \{I, X, Y, Z\}$$

* by context will use this set as operators or labels

Orthogonal & complete set

$$\langle P_a, P_b \rangle = 2^n \delta_{ab}$$

$$\langle P_a, P_b \rangle = \text{Tr} (P_a^\dagger P_b)$$

(for all a, b in the set)

Example decomposition

of a qubit mixed state in terms of Paulis

$$\rho = \frac{1}{2} (I + r_X X + r_Y Y + r_Z Z)$$

Pauli decomposition of a mixed state (holds for qubits)

$$\rho = \sum_{a \in \mathbf{P}} r_a P_a$$

linear vector decomposition
onto orthogonal basis

$$r_a = \frac{\langle P_a, \rho \rangle}{\langle P_a, P_a \rangle}$$

Inner product of Hermitian
operators $r_a \in \mathbb{R}$

Action of single-qubit Pauli gate on mixed state basis

Pauli gate on a mixed state: conjugation of ρ by Pauli

$$\rho \text{ --- } \boxed{P_a} \text{ --- } \mathcal{P}_a \rho = P_a \rho P_a^\dagger$$

Pauli gate

Conjugation map

		Density matrix component			
		I	X	Y	Z
$\mathcal{P}_a(P_b)$					
$I \cdot I$		I	X	Y	Z

Basis element by element
linear map

$$\begin{aligned} \mathcal{P}_a(\rho) &= \sum_b r_b \mathcal{P}_a(P_b) \\ &= \sum_b r_b P_a P_b P_a \end{aligned}$$

(for the experts in the audience,
using Z_2^2 representation)

$$= (-1)^{\langle a, b \rangle_{\text{Sp}}} r_b P_b$$

Action of single-qubit Pauli gate on mixed state basis

Pauli gate on a mixed state: conjugation of ρ by Pauli

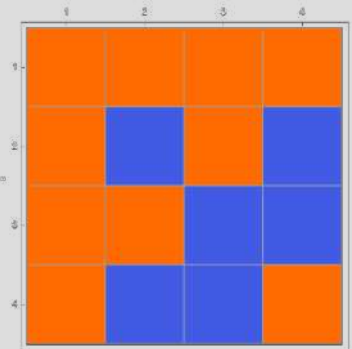
$$\rho \xrightarrow{\text{Pauli gate } P_a} \mathcal{P}_a \rho = P_a \rho P_a^\dagger$$

Walsh-Hadamard transform

$$H_2 = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

*caution: ordering

matrix
plot

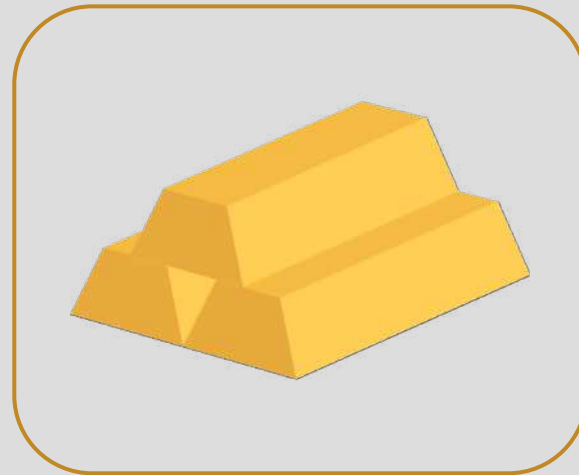


- equivalent to a multidimensional DFT of size 2^n
- +1, -1 eigenvalues
- an orthogonal, symmetric, involutive, linear operation on 2^n real numbers
- note relation of our matrix to symplectic product Z_2^2 representation

Conjugation map

Conjugation map			Density matrix component				
			$\mathcal{P}_a (P_b)$	I	X	Y	Z
Gate	I	$I \cdot I$	I	X	Y	Z	
	X	$X \cdot X$	I	X	$-Y$	$-Z$	
	Y	$Y \cdot Y$	I	$-X$	Y	$-Z$	
	Z	$Z \cdot Z$	I	$-X$	$-Y$	Z	

Pauli's are Gold!

 P_a $=$ 

Action of single-qubit Pauli gate on mixed state basis

Pauli gate on a mixed state: conjugation of ρ by Pauli

$$\rho \text{ --- } \boxed{P_a} \text{ --- } \mathcal{P}_a \rho = P_a \rho P_a^\dagger$$

Pauli gate

Conjugation map

Conjugation map			Density matrix component				
			$\mathcal{P}_a(P_b)$	I	X	Y	Z
Gate	I	$I \cdot I$	I	X	Y	Z	
	X	$X \cdot X$	I	X	$-Y$	$-Z$	
	Y	$Y \cdot Y$	I	$-X$	Y	$-Z$	
	Z	$Z \cdot Z$	I	$-X$	$-Y$	Z	

Superoperator lens

$$\rho \mapsto |\rho\rangle\rangle \quad \text{vec}$$

$$P_a \mapsto |P_a\rangle\rangle \quad \text{vec}$$

$$P_a \cdot P_a^\dagger \mapsto \mathcal{P}_a \quad \text{op}$$

$$P_a \rho P_a^\dagger \mapsto \mathcal{P}_a |\rho\rangle\rangle$$

Key vectorization identity (row stacking)

$$\text{vec}(A_0 B A_1^\top) = (A_0 \otimes A_1) \text{vec}(B)$$

$$\text{vec}(P_a \rho P_a^\dagger) = (P_a \otimes P_a^*) \text{vec}(\rho)$$

Use as basis elements of $\text{Op}(H)$ and $\text{Op}(\text{Op}(H))$

$$\text{Tr}(P_a^\dagger \cdot) = \langle\langle P_a | \cdot$$

$$P_a \text{Tr}(P_b^\dagger \cdot) = |P_a\rangle\rangle \langle\langle P_b | \cdot$$

Superoperator Pauli transfer matrix representation

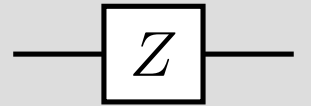
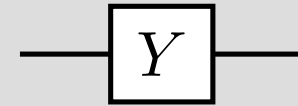
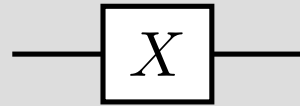
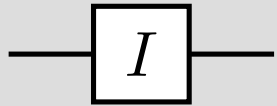
Pauli gate on a mixed state: conjugation of ρ by Pauli

$$\rho \text{ --- } \boxed{P_a} \text{ --- } \mathcal{P}_a \rho = P_a \rho P_a^\dagger$$

Pauli superoperator

$$\mathcal{P}_a = \sum_b (-1)^{\langle a, b \rangle_{\text{Sp}}} |P_b\rangle\rangle \langle\langle P_b|$$

Pauli transfer matrix: chi matrix in the Pauli basis



$$\begin{array}{l} \mathcal{I}: \begin{array}{c} I \\ Z \\ X \\ Y \end{array} \begin{pmatrix} I & Z & X & Y \\ 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix} \quad \mathcal{Z}: \begin{pmatrix} I & Z & X & Y \\ 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix} \quad \mathcal{X}: \begin{pmatrix} I & Z & X & Y \\ 1 & & & \\ & -1 & & \\ & & 1 & \\ & & & -1 \end{pmatrix} \quad \mathcal{Y}: \begin{pmatrix} I & Z & X & Y \\ 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & 1 \end{pmatrix} \end{array}$$

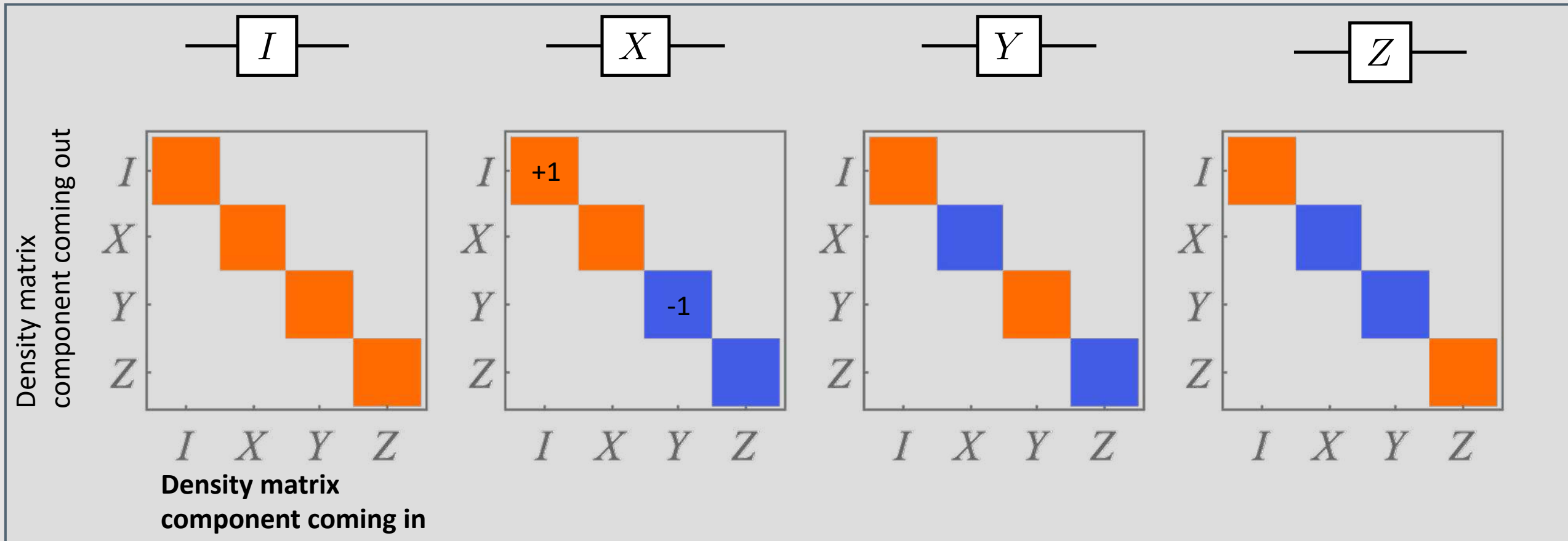
*caution: ordering is based on binary Z_2^2 notation

Visualizing the PTM

Pauli gate on a mixed state: conjugation of ρ by Pauli

$$\rho \longrightarrow \boxed{P_a} \longrightarrow \mathcal{P}_a \rho = P_a \rho P_a^\dagger$$

- diagonals are just columns of the WH matrix





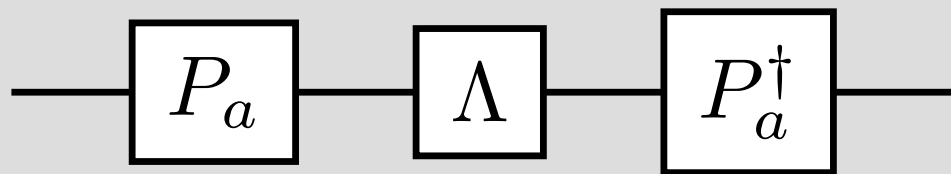
* pininterest

Twirl channel with Pauls

(conjugating a channel by a Pauli gate)

Twirl: acting on super operators

Super-super operators



Algebraic expression of channel sequence:

$$\mathcal{P}_a \Lambda \mathcal{P}_a^\dagger = P_a^\dagger \Lambda (P_a \cdot P_a^\dagger) P_a$$

↙
Vectorize middle channel



Key vectorization identity (row stacking)

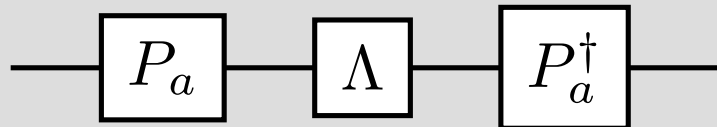
$$\text{vec}(A_0 B A_1^T) = (A_0 \otimes A_1) \text{vec}(B)$$

on basis elements

$$|P_a\rangle\rangle\langle\langle P_b| \cdot \mapsto |P_a, P_b\rangle\rangle\rangle$$

Twirl: acting on super operators

Super-super operators



Single qubits

16 basis elements for superoperators:

$I, IX, IY, IZ, XX, XY, \dots$



I



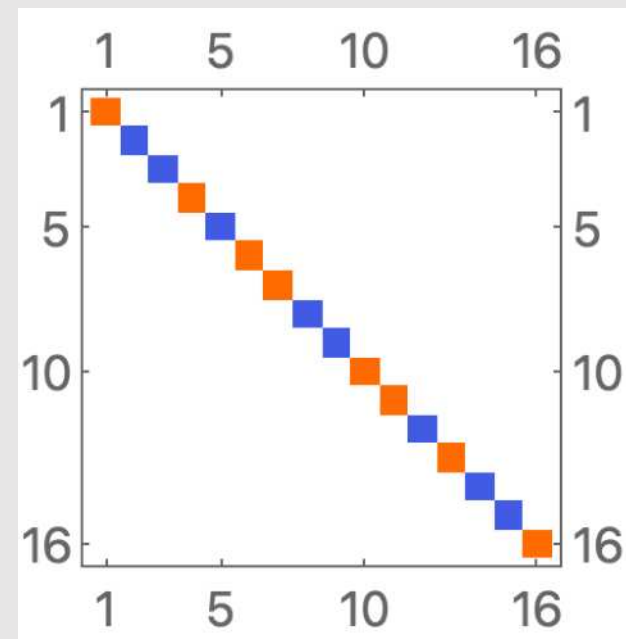
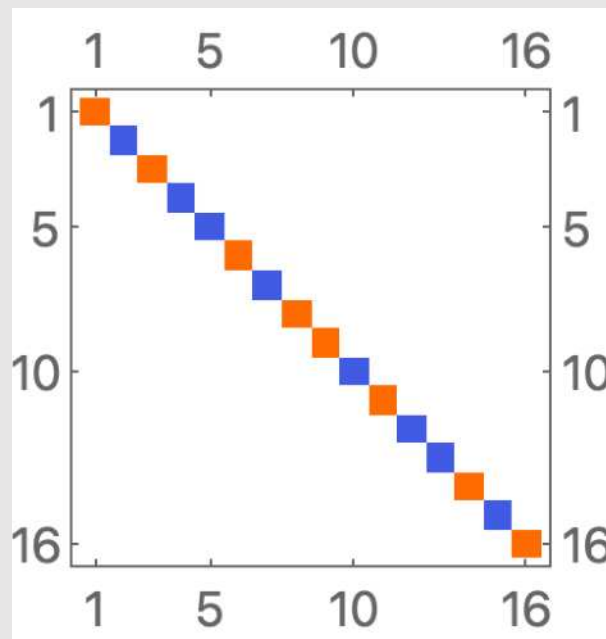
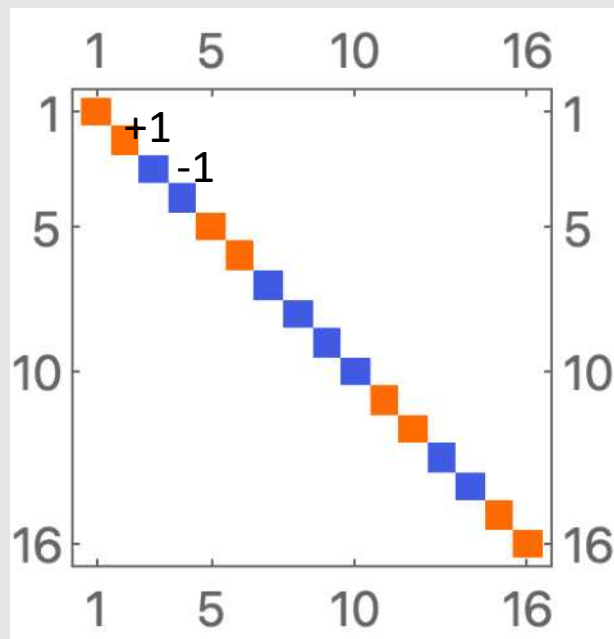
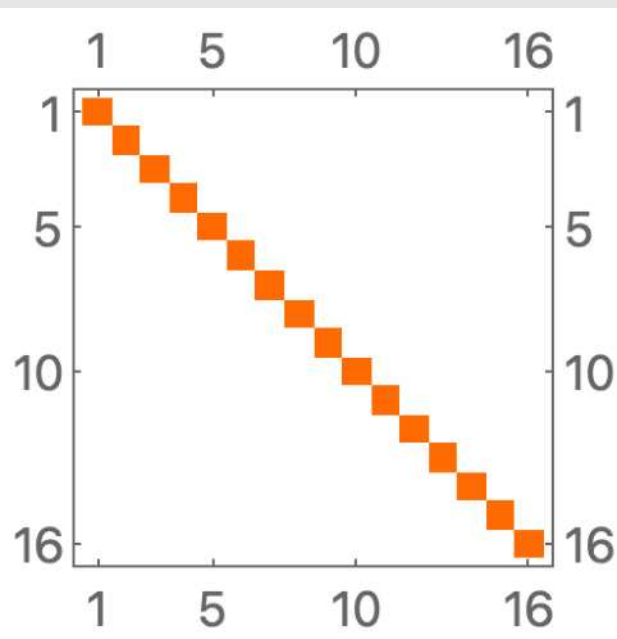
X



Y



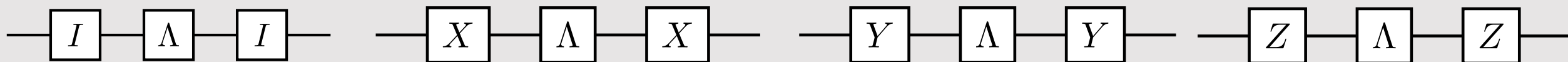
Z



Twirl: acting on super operators

$$\text{---} \boxed{P_a} \text{---} \boxed{\Lambda} \text{---} \boxed{P_a^\dagger} \text{---} \mathcal{P}_a \Lambda \mathcal{P}_a^\dagger$$

Super super operators

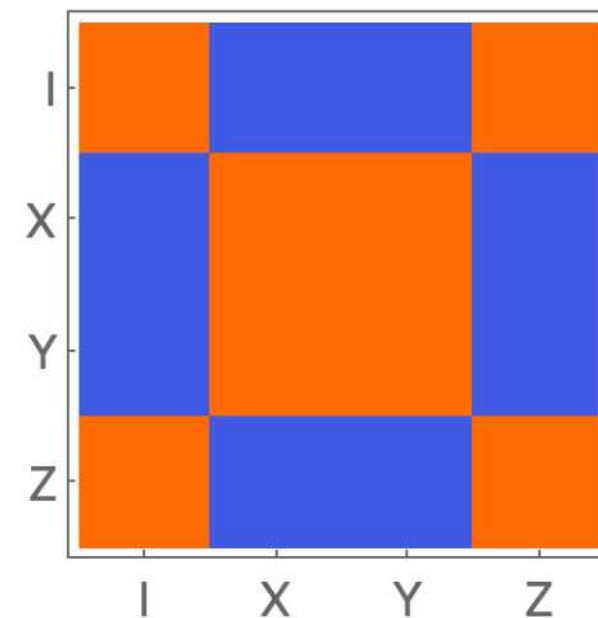
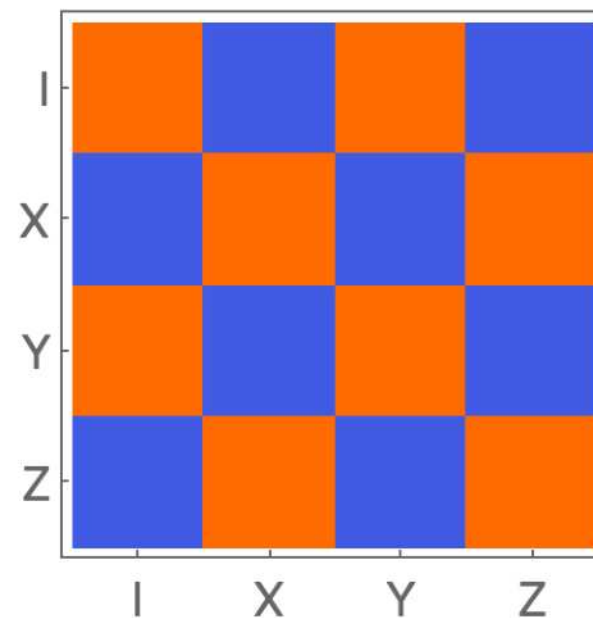
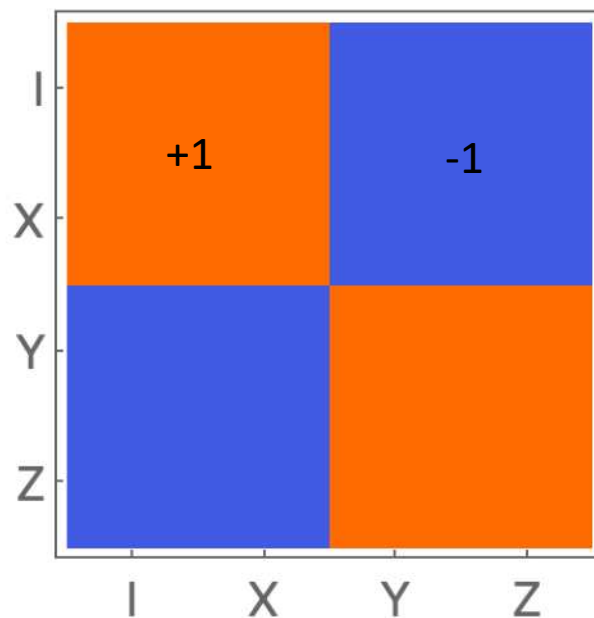
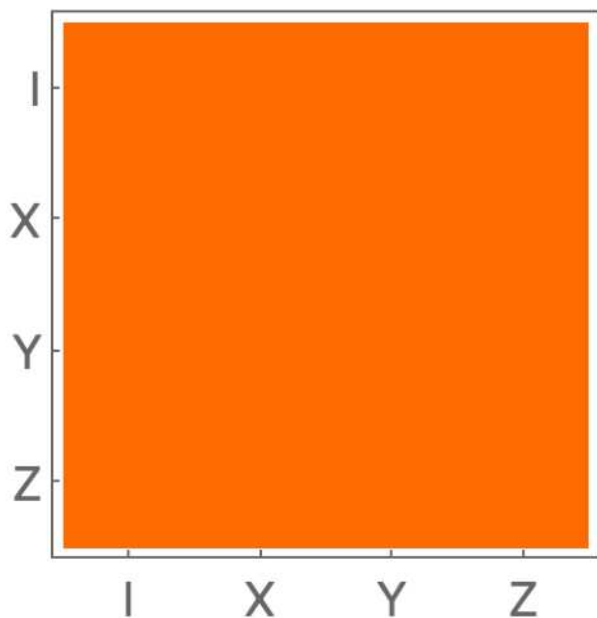


I

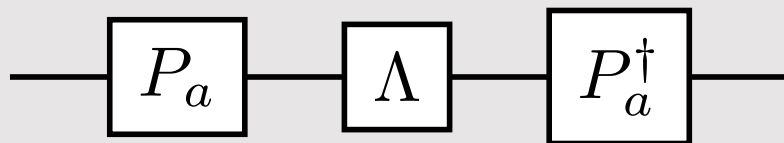
X

Y

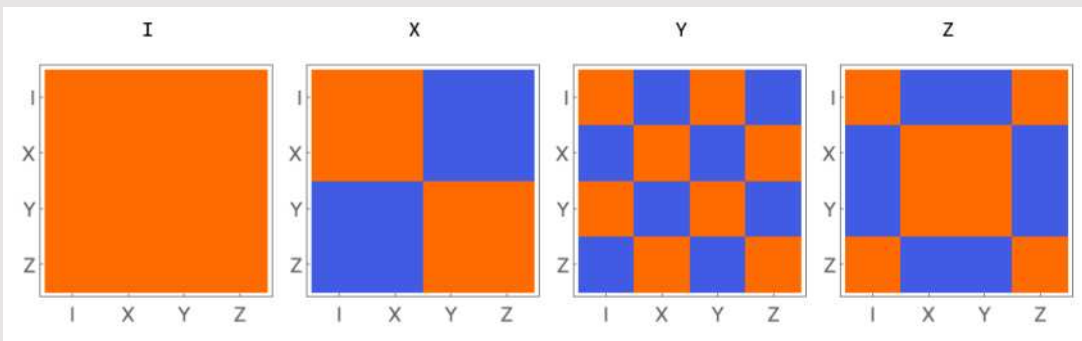
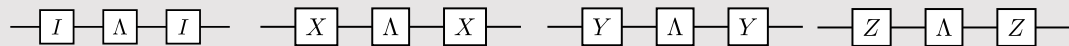
Z



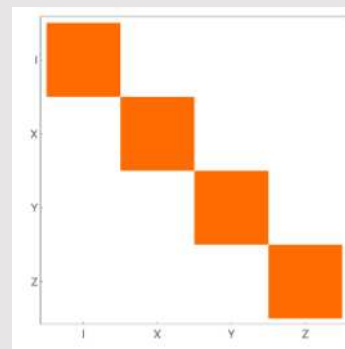
Twirl: acting on super operators



Average over these masks $\mathcal{M}(P_a)$



=



Directly **MASK** elements of superoperator Lambda in PTM basis

$$\frac{1}{|\Sigma|} \sum_{a \in \Sigma} \mathcal{P}_a \Lambda \mathcal{P}_a^\dagger = \left(\frac{1}{|\Sigma|} \sum_{a \in \Sigma} \mathcal{M}(\mathcal{P}_a) \right) \odot \Lambda = \mathcal{M} \odot \Lambda$$

↑
element-wise product (Hadamard)

Designs